

Ultrahigh resolution real time OCT imaging using a compact femtosecond Nd:Glass laser and nonlinear fiber

S. Bourquin, A. D. Aguirre, I. Hartl, P. Hsiung, T. H. Ko and J. G. Fujimoto

Department of Electrical Engineering and Computer Science and Research Laboratory of Electronics,
Massachusetts Institute of Technology, Cambridge, MA 02139, USA
bourquin@mit.edu

T. A. Birks and W. J. Wadsworth

Optoelectronics Group, Department of Physics
University of Bath
Bath, BA2 7AY, United Kingdom

U. Bunting and D. Kopf

High Q Laser Production GmbH
A-6845 Hohenems, Austria

Abstract: Ultrahigh resolution, real time OCT imaging is demonstrated using a compact femtosecond Nd:Glass laser that is spectrally broadened in a high numerical aperture single mode fiber. A reflective grating phase delay scanner enables broad bandwidth, high-speed group delay scanning. We demonstrate *in vivo*, ultrahigh resolution, real time OCT imaging at 1 μm center wavelength with <5 μm axial resolution in free space (<4 μm in tissue). The light source is robust, portable, and well suited for *in vivo* imaging studies.

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1. Introduction

Optical coherence tomography (OCT) is an emerging technology for micron-scale cross-sectional imaging of biological tissue and materials [1-3]. OCT is analogous to ultrasound except that it uses low coherence light instead of sound to perform ranging or imaging. High resolution detection of reflected or backscattered light is performed using low coherence interferometry. One of the key limitations in OCT has been the lack of compact, high performance, low coherence light sources with sufficient bandwidth and power to enable high resolution, real time imaging. High nonlinearity, air-silica microstructure fibers [4] or tapered fibers [5] can generate broadband continuum that spans the visible to the near infrared using femtosecond pulses. These fibers have enhanced nonlinearity because of dispersion characteristics of these fibers, which shift the zero dispersion to shorter wavelengths, and the small core diameters, which provide tight mode confinement. High numerical aperture fibers also have been used with femtosecond Ti:Sapphire lasers to achieve bandwidths of up to 200 nm [6]. Continuum generation from a femtosecond Ti:Sapphire laser with air-silica microstructured photonic crystal fibers was demonstrated to achieve OCT image resolutions of 2.5 μm in the spectral region 1.2 μm to 1.5 μm [7], resolutions of 1.3 μm in the spectral region 800 nm to 1400 nm [8], and resolution <1 μm in the spectral region of 550 nm to 950 nm [9]. While high resolution imaging was demonstrated, femtosecond Ti:Sapphire lasers are generally expensive and bulky. Previous studies also used galvanometer or voice coil scanned retroreflector delay lines which had limited axial scan speed, so that real time imaging was not performed.

In this paper, we demonstrate a new broadband light source for ultrahigh resolution OCT imaging based on a compact, commercially available Nd:Glass femtosecond laser that is spectrally broadened in a high numerical aperture single mode fiber. We also describe a reflective grating phase delay line design that enables high speed, broad bandwidth optical delay scanning. Ultrahigh resolution, high speed OCT imaging is achieved at 1 μm center wavelength with <5 μm axial resolution in free space (or $\sim$ 4 μm in tissue) and 93 dB sensitivity. Axial scan speeds of $\sim$ 11 m/s are demonstrated, which is sufficient for real time

imaging. This system is compact and well suited for *in vivo* ultrahigh resolution OCT imaging studies. The 1 μm wavelength is important because it has improved image penetration depths when compared to 800 nm and ultrahigh resolution imaging requires less bandwidth than at 1.3 μm wavelengths [10].

2. Experimental setup

Figure 1 shows a schematic of the experimental setup. A compact, commercially available diode pumped femtosecond Nd:Glass laser (High Q Laser Production) generates pulses at 1064 nm with 100-150 fs duration, 145 mW average power and 75 MHz repetition rate. The laser used for this study measured only 53 cm x 20 cm, and even smaller lasers, measuring 18 cm x 10 cm will soon be available. The Nd:Glass laser is pumped by two 1 W laser diodes and is soliton modelocked [11] using a semiconductor saturable absorber mirror (SESAM) [12,13] for self starting and intracavity prisms for dispersion compensation. To generate a continuum centered at 1 μm wavelengths, the femtosecond pulses from the Nd:Glass laser are coupled into a 2 meter length of commercially available high numerical aperture (NA) single mode fiber with a 2.75 mm focal length aspheric AR coated lens. The fiber has a germanium doped core with a 2.5 μm mode field diameter to provide enhanced nonlinear effects. The fiber dispersion was approximately -125 ps/nm/km. Other types of nonlinear optical fibers also can be used. For example, a tapered fiber consisting of a standard single mode fiber tapered down to a uniform waist with a diameter of 2 μm can perform efficient continuum generation [5]. Ultrahigh resolution OCT imaging with <5 μm resolution in the 1.3 μm range recently has been demonstrated using this method [14].

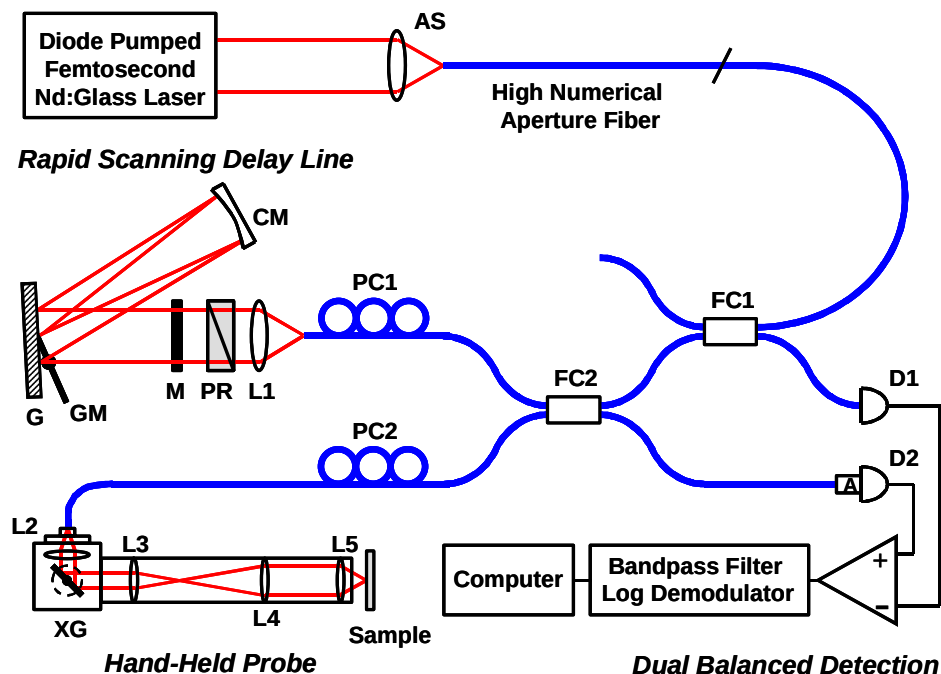


Fig. 1. Ultrahigh resolution, real time, OCT system using a femtosecond diode pumped Nd:Glass laser with a high numerical aperture fiber and a broadband grating based phase delay scanner. AS: aspheric lens, NA: numerical aperture, L1-L5: chromatically corrected doublet, FC1,2: 3dB 2x2 fiber coupler, PC1,2: polarization controller, A: air gap, D1,2: InGaAs photodiode, PR: prisms, M: mirror, DM: mirror for double-pass configuration, G: grating, CM: curved mirror, GM: galvanometer controlled mirror, XG: x-galvanometer.

The OCT system consists of a dual-balanced interferometer with two commercially available 50/50 fiber couplers (FC1 and FC2) at 1064 nm center wavelengths supporting more

than 150 nm of bandwidth. High speed group delay scanning is performed using a grating-based phase delay scanner [15], which has all-reflective optics. The sample arm contains a hand-held probe for *in vivo* imaging with a transverse galvanometer scanner. Dispersion is matched in the two arms of the interferometer in order to maintain resolution and fringe contrast in the interference signal. Dispersion matching was performed by matching the fiber lengths and matching lens materials used in the sample arm by identical material and thickness glass prisms (PR) in the reference arm. Since different glass materials have their zeros of dispersions at different wavelengths, this method ensures that both second- and third-order variations in dispersion are matched in the sample and reference arms. Dispersion mismatch can be monitored by taking the phase of the Fourier transform of the interferometer fringe signal. Polarization controllers (PC) are used in both sample and reference arms.

Dual-balanced InGaAs photodiodes (D1, D2) are used to cancel excess laser noise. The interferometric signal occurs 180 degrees out of phase in the two detectors, therefore subtracting the photodiode signals adds the signal of interest, but cancels the excess noise from the laser source. The input to photodiode D2 is attenuated by approximately 3 dB in order to account for the interferometric signal measured at photodiode D1 that is attenuated by 3 dB on its backward pass through the coupler (FC1).

In order to perform high speed, broad bandwidth delay scanning, a new reflective phase delay grating scanner was developed. The use of reflective optics avoids the need for dispersive large diameter lenses that are typically required in phase delay scanners. The reflective design also improves bandwidth because chromatic aberrations are reduced. Figure 2 shows a top view and a side view of the reflective phase delay scanner. The grating phase delay scanner is similar to diffraction grating-based systems used in ultrafast optics for pulse shaping and chirped-pulse amplification [16,17]. The delay line uses a grating and curved mirror to produce spectral dispersion. A mirror in the Fourier plane is angle scanned by a high speed resonant galvanometer (EO Technologies). This produces a phase shift in the Fourier plane, resulting in a group or phase delay in the time domain. Depending upon the offset of the center of rotation of the mirror, the group and phase delay can be scanned at different rates [15]. The group delay dispersion can also be tuned by displacing the diffraction grating from the focal plane of the mirror.

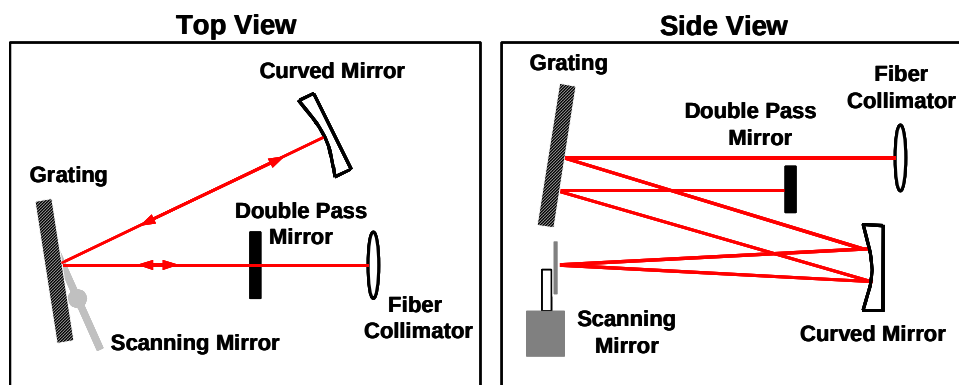


Fig. 2. Top and side views of the reflective grating phase delay line for high speed imaging. The double pass mirror allows doubling the axial scan range and coupling back the light beam into the fiber.

The delay line uses a 300 g/mm plane reflective grating and a 5 cm focal length curved mirror placed one focal length from the grating. The collimated spectrum is scanned using a 12 mm wide galvanometer mirror with a 1000 Hz sine waveform, thus enabling 2000 axial scan lines to be acquired per second. This scan speed enables 500 pixel images to be acquired at 4 frames per second or 250 pixel images to be acquired at 8 frames per second. The double

pass geometry doubles the scan range and cancels the effects of spectral offset in the beam after a single pass, thus enabling more efficient recoupling of the light into the fiber. The grating phase delay scanner enables group and phase delays to be independently controlled by offsetting the transverse position of the galvanometer mirror [15]. A pure phase modulation, without group delay scanning also can be achieved by using grating phase delay scanners [18]. The group velocity was ~ 11 m/s and the linear group delay scan range was ~ 3 mm in depth. The phase delay scan velocity was set to produce a Doppler frequency of 5 MHz by offsetting the center of rotation of the galvanometer mirror. The photodetector output was bandpass filtered at a center frequency of 5 MHz with a 2.7 MHz bandwidth.

3. Results and discussion

3.1 System performance

Figure 3(a) shows a typical spectrum generated by the high numerical aperture fiber with 90 mW of average output power and a bandwidth of 139 nm. This bandwidth should yield a theoretical axial resolution of $3.6 \mu\text{m}$. The nearly symmetric broadening of the spectrum is the result of self phase modulation in the normal, positive dispersion operating regime. Broader spectra can be generated by using other nonlinear fibers, such as photonic crystal fibers, which operate near zero dispersion. However, the processes that generate very broad continua in this regime are the result of a combination of self phase modulation, Raman, soliton, and four-wave mixing effects. These are often associated with strong spectral modulation and high excess noise [19]. Operation in the normal dispersion regime is dominated by self phase modulation and chromatic dispersion, which typically yield smoother spectra.

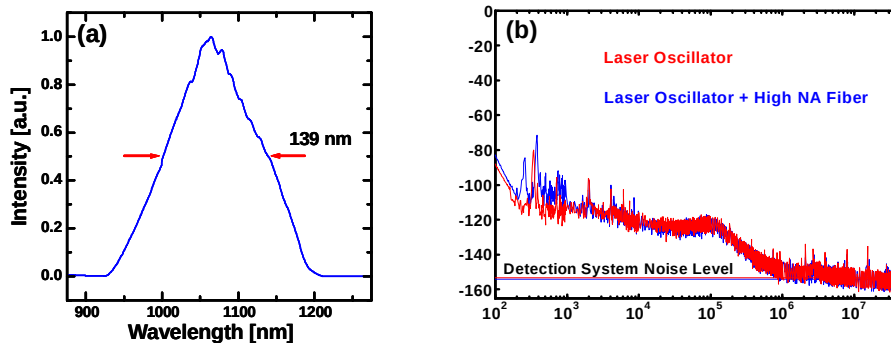


Fig. 3. (a) Plot of the optical spectrum of the Nd:Glass laser broadened in the high numerical aperture (NA) single mode fiber. (b) Comparison of RF noise spectrum for the Nd:Glass pump laser to that of the continuum generated in the high NA fiber.

The intensity noise was measured using a fast photodetector and an RF spectrum analyzer. Figure 3(b) compares the RF noise spectrum of the femtosecond Nd:Glass laser output to that of the continuum generated from the high NA fiber. The noise measurements are performed at the first harmonic of the repetition rate of the mode-locked femtosecond laser. This measurement method normalizes the noise power to the carrier frequency power and provides a calibrated measure of the noise. The detected carrier power was 1.3 mW for the laser oscillator noise measurement and 1.6 mW for the continuum generated from the high NA fiber. The excess noise of the laser decreases rapidly at frequencies above 10^5 Hz, which is expected for frequencies above the relaxation oscillation resonance, where laser intensity noise is reduced by integration effects. No increase in excess noise above the existing noise in the femtosecond Nd:Glass output is observed. This contrasts with measurements we have performed of broadband continua generated in photonic crystal fibers, which can have 20 to

30 dB greater RF intensity noise than the pump laser alone. These measurements also suggest that imaging using high frequency demodulation in the MHz range is desirable because there is less total RF noise that must be cancelled by the dual-detector interferometer.

The sensitivity of the system was measured to be 93 dB, with an optical power on sample of 14 mW. This sensitivity can be improved if a linear versus a sinusoidal group delay scan is used, since this would enable the detection bandwidth to be reduced [20]. There was also approximately 3 dB loss in the handheld scanning probe galvanometers and relay lenses, which could be reduced with an improved design. Finally, it is interesting to note that while the dual-balanced interferometer configuration shown here has the advantage of simplicity, there is an additional 3 dB loss in power transmission from the source to the sample compared to a single coupler configuration. Furthermore, the efficiency of signal detection from the sample is the same as in a single coupler interferometer because, although two detectors are used, they are both attenuated by 3 dB, when compared to a single coupler interferometer. The system efficiency could be significantly improved if circulators were available at 1 μm , because the power transmission efficiency from the source to the sample could be increased by 3 dB and the signal collection efficiency from the sample to the detectors could be increased by 3 dB by using a circulator place of the coupler (FC1) [21].

The system resolution and point spread function was characterized by measuring reflections from a 150 μm thick microscope cover glass (optical thickness 225 μm). To avoid saturation of the detector, the sample arm was attenuated using a 1.5 OD neutral density filter, which corresponds to a 3.0 OD round trip attenuation. Figure 4(a) and (b) shows the interference signal, with an axial resolution of 4.5 to 5 μm in air (corresponding to <4 μm in tissue). The difference from the theoretically predicted resolution values may be due to imperfect matching of the dispersion between the arms of the interferometer as well as the non-Gaussian shape of the spectrum.

Figure 4(c) shows the log demodulated signal. The point spread function has excellent behavior on the log scale with no pronounced sidelobes or wings. This is consistent with the smooth spectrum and low noise shown in Figure 3(a) and 3(b). Broader bandwidths and narrower point spread functions can be generated using nonlinear fibers, such as photonic crystal fibers, operating near zero dispersion. However these broad spectra are often the result of averaging fluctuating, highly modulated spectra. Since the quality of the interferometric point spread function is determined by the smoothness of the spectrum, a modulated spectrum produces sidelobes or wings in the point spread function. Even if the spectrum appears smooth because fluctuations are averaged, the sidelobe or wings in the point spread function are not reduced on the log scale by averaging over spectral fluctuations. Since OCT imaging is typically performed using a log grey or false color scale, a clean and well-behaved point spread function on a log scale is very important for good image quality.

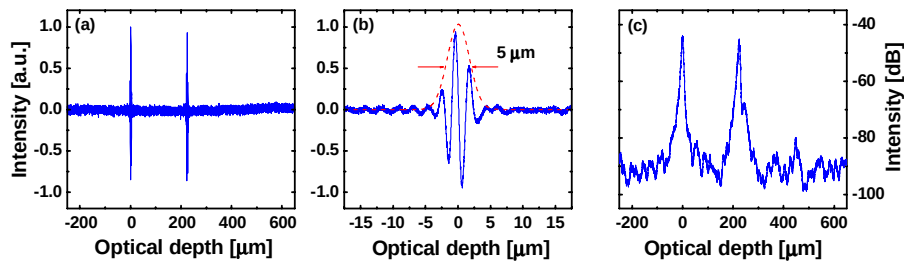


Fig. 4. (a) Plot of the interference signal and (b) measurement of the axial resolution using a 150 μm thick (225 μm optical thickness) microscope cover glass test sample. (c) Corresponding log demodulated signal. A 1.5 ND filter was inserted in the sample arm to avoid detector saturation.

3.2 *In vivo* imaging measurements

The utility of the system for performing ultrahigh resolution, real time imaging *in vivo* was demonstrated by imaging the hamster cheek pouch and human skin. Imaging was performed using a compact handheld scanning probe. Figures 5(a) and (b) show the cheek pouch of a Syrian golden hamster. This tissue was chosen because its morphology is similar to squamous epithelial structure in human tissues. A junction between two blood vessels is visible in the picture. Imaging of human skin was also demonstrated. Figure 5(c) shows a region of the volar fingertip, where the stratum corneum and sweat ducts can be clearly resolved. Figure 5(d) shows the fingernail bed, where the dermal-epidermal junction can be differentiated. All images were acquired at 4 frames per second, are 2.5 mm x 1.2 mm in size (transverse x axial), and contain 500 x 750 pixels. The transverse resolution is 11 μm , and the axial resolution is <4 μm in tissue.

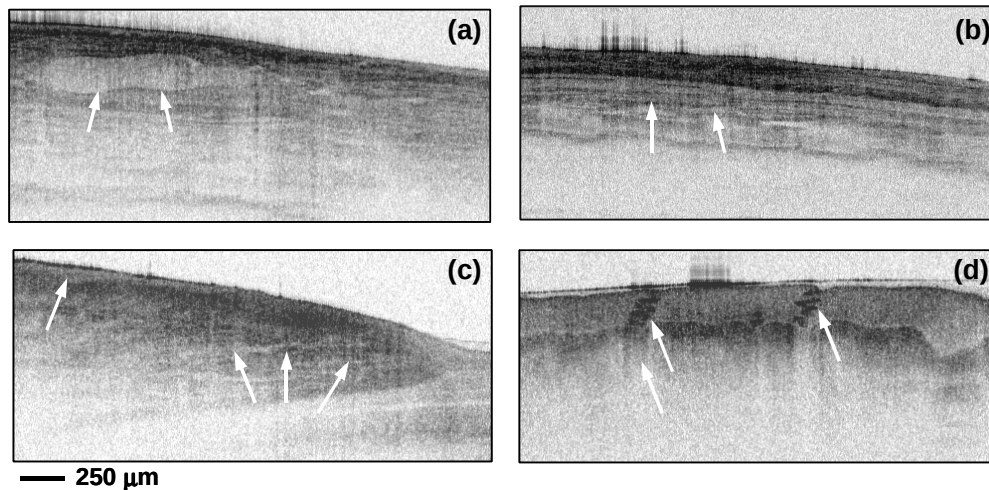


Fig. 5. *In vivo* ultrahigh resolution OCT images at 1 μm wavelength. Images were acquired at 4 frames per second and are 2.5 mm transverse x 1.2 mm axial with 11 μm x 4 μm resolution. (a,b) Images of the hamster cheek pouch showing squamous epithelium and connective tissue as well as a junction between two vessels. (c) Image of the nail bed showing the stratum corneum and the junction between the epidermis and the dermis. (d) Image of human volar finger pad showing stratum corneum and sweat ducts.

To demonstrate the real time imaging capability of the system, sequences of two-dimensional cross sectional images were recorded at a rate of 4 Hz. Figure 6(a) shows an example of a movie of human volar finger pad with sweat ducts in the stratum corneum. The field of view is 1.9 mm x 1.3 mm. The MOV file shows the natural movement of the skin and corresponding changes in the cross sectional view of the sweat ducts. Figure 6(b) shows an example of real time imaging in multiple transverse planes at different positions scanning from the posterior to the anterior of a *Xenopus laevis* tadpole. The field of view is 1.8 mm x 1.5 mm. The images, taken through the region of the head, clearly show the eyes and the neural tube.

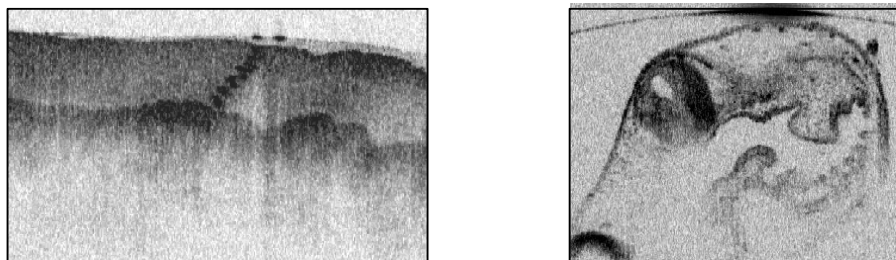


Fig. 6. Movie from *in vivo* ultrahigh resolution images acquired at 4 frames per second. (a) (2.1 MB) 2D images of human volar finger pad with sweat ducts (1.9 mm x 1.3 mm). (b) (2.3 MB) Sequence of transverse images of a tadpole in water scanned from the posterior to the anterior (1.8 mm x 1.5 mm).

4. Conclusion

In summary, we have demonstrated a new, compact high performance light source for high speed, ultrahigh resolution OCT imaging using a diode pumped femtosecond Nd:Glass laser that is spectrally broadened in a high numerical aperture optical fiber. We also demonstrated a new reflective phase delay scanner that achieves broad bandwidth, high speed delay scanning. We achieved ultrahigh resolution, real time imaging with $<4\ \mu\text{m}$ axial resolution in tissue. The sensitivity of the system can be further improved by several dB if a circulator can be used in the dual-balanced interferometer and if the phase scanning can be made more linear in order to reduce excess detection bandwidth. With further improvements in nonlinear fibers, even broader bandwidths and finer axial resolutions should be achievable. Since this system uses a commercially available femtosecond laser (High Q Laser Production), this study promises wider availability of ultrahigh resolution, high speed OCT imaging. The system is compact and robust, it achieves ultrahigh resolution, real time imaging, and it promises a wider range of imaging studies.

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